



COVENANT UNIVERSITY

CANAANLAND, KM 10, IDIROKO ROAD
P.M.B 1023, OTA, OGUN STATE, NIGERIA.

B.Sc. DEGREE EXAMINATION

COLLEGE: Science & Technology

DEPARTMENT: Computer & Info. Science

SESSION: 2023/2024

SEMESTER: Alpha

COURSE CODE: CSC 214

CREDIT UNIT: 3

COURSE TITLE: High Performance Computing & Database Mgt.

TIME: 3 HOURS

INSTRUCTIONS: Attempt any four (4) questions and write legibly.

Question 1

(a) Define the following relational concepts:

- (i) Superkey (1 mark)
- (ii) Primary key (1 mark)
- (iii) Foreign key (1 mark)
- (iv) Candidate key (1 mark)

(b) Use the EmployeeInfo relational database schema to answer questions (i-vii)

EmployeeInfo Schema

Emp (Eid, Ename, Age, Salary)

Works (Eid, Did, per_time)

Dept (Did, Dname, Budget, managerId)

- (i) Write an SQL statement to create the relations in the EmployeeInfo schema (4½ marks)
- (ii) Write an SQL statement to add a 40 years Azi Esther whose employee_id is 101, and with salary of 15000. (1½marks)
- (iii) Write an SQL statement to give every employee a 10 percent salary increase. (1½ marks)
- (iv) Write an SQL statement to delete the Toy department. (1½mark)
- (v) Write an SQL query to increment the budget of the account department to 8 million. (1½marks)
- (vi) Write an SQL statement to remove the Dept relation completely from the EmployeeInfo Database (1½marks)
- (vii) Write an SQL statement to make the column Eid come first when the Emp relation is queried. (1½marks)

Question Two

- (a) Differentiate between the following relational concepts with Example (2½marks)
 - (i) Strong and weak entity (2½marks)
 - (ii) Disjoint and Overlap participation
- (b) Business rule is important in data modelling, as it represent an organisation operation interaction among the stakeholders and the data flow. As a database administrator you have been asked by the management of Alex & Suite legal firm to come up with a standard business rule that will capture their organisation operation interaction and data flow. Explain Four important qualities you will need to consider in making your delivery. (4marks)

- (c) Each case handled by Alex & Suite firm has a unique case number; a date opened, date closed, and judgment description are also kept on each case. A case is brought by one or more plaintiffs, and the same plaintiff is involved in at least one case. A plaintiff has a requested judgment characteristic. A case is against one or more defendants, and the same defendant is usually involved in one or many cases. A plaintiff or defendant may be a person or an organization usually referred to as a legal entity. Over time, the same person or organization may be a defendant or a plaintiff in cases. In either situation, such legal entities are identified by an entity number, and other attributes are name and net worth.

Develop an ERD/EERD for the business rule follow good data naming guidelines and constraints.

(8½marks)

Question Three

- (a) What is Normalization (1½marks)
 (b) Explain the following Terms with example (1½marks)
 (i) Full Functional Dependency (1½marks)
 (ii) Partial Functional Dependency (1½marks)
 (iii) Transitive functional Dependency (1½marks)
 (c) Given the **Customer_renter Relation** and the information below answer the following question (2marks)
 (i) Draw the functional dependency diagram for the relation (2marks)
 (ii) Normalize the relation to 3rd Normal forms. The primary key is Customer_No, Property_No. (9½marks)

Customer_renter Relation

Customer_No	CName	Property_No	Address	RentStart	RentFinish	Rent	Owner_No	OName
CR76	John Kay	PG4	6, Lawrence St. Ikoyi	1-Jul-04	31-Aug-06	350	CO40	Tina Murphy
		PG16	5, Novar St, Oshodi	1-Sep-06	1-Sep-08	450	CO93	Tony Shaw
CR36	Fred Wey	PG4	6, Lawrence St. Ikoyi	1-Sep-02	10-Jun-04	350	CO40	Tina Murphy
		PG36	2, Fred St. Apapa	10-Oct-04	1-Dec-06	375	CO93	Tony Shaw
		PG16	5, Novar St, Oshodi	1-Jan-06	10-Aug-06	450	CO93	Tony Shaw

Functional dependencies:

- fd1: Customer_No, Property_No → RentStart, RentFinish (Pk)
 fd2: Customer_No → CName (Pd)
 fd3: Property_No → Address, Rent, Owner_No, OName (Pd)
 fd4: Owner_No → OName (Td)
 fd5: Customer_No, RentStart → Property_No, Address, Rent, Owner_No, OName (Ck)
 fd6: Property_No, RentStart → Customer_No, CName, RentFinish (Ck)

Question Four

- (a) Briefly differentiate the following pair of concepts with example
 (i) Supertypes and subtypes
 (ii) Generalization and specialization

- (b) Transform the conceptual design for patient and treatment in Figure 1 into a logical design
 (2½ marks)
 (2½ marks)
 (6 marks)

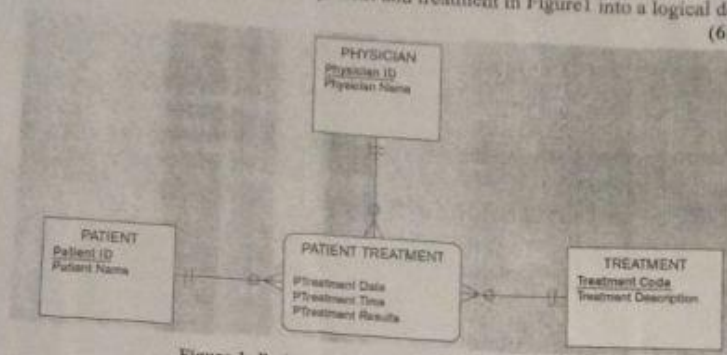


Figure 1: Patient treatment conceptual design

- (c) Giving an EER diagram for Vacation Property Rentals in Figure 2 Transform the EER diagram to a set of relations and develop its relational/logical schema.
 (6½ marks)

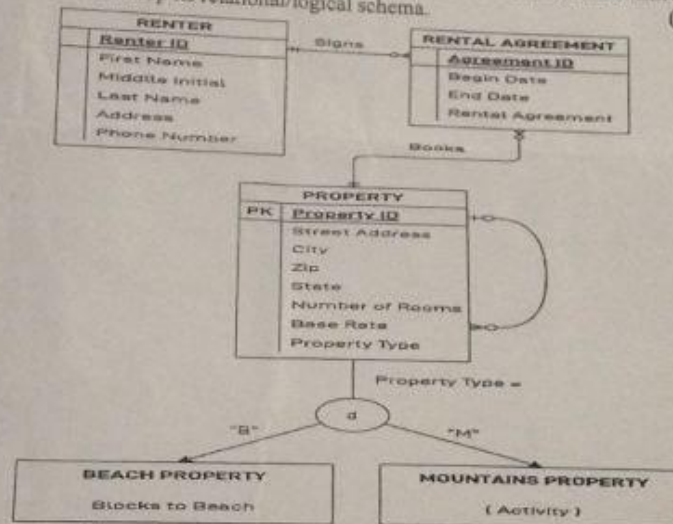


Figure 2: EER diagram for Vacation Property Rentals

Question Five

- (a) State four justifications for organizations to adopt the conversion from the traditional file processing system to the database approach. (2 marks)
 (b) Explain two (2) costs and risks of adopting the database approach in managing data for a business that previously used a traditional file processing system. (2 marks)
 (c) State and describe three integrity controls to be considered in designing a data field. (4½ marks)
 (d) Define Vertical and horizontal partitioning (3 marks)

- (e) Explain four benefits horizontal partitioning (4 marks)
 (f) Define the following terms under Hash file organization (2 marks)
 (i) Hash Index Table
 (ii) Pointer
 (iii) Hashed file organization
 (iv) Hashing algorithm

Question Six

- (a) What is an attribute? Explain four different types of attributes with example. (4 marks)
 (b) Based on the Figure 3, answer the following questions and explain the rationale for your response. For each question, identify the element(s) in the E-R diagram that you used to determine your answer. (13 marks)

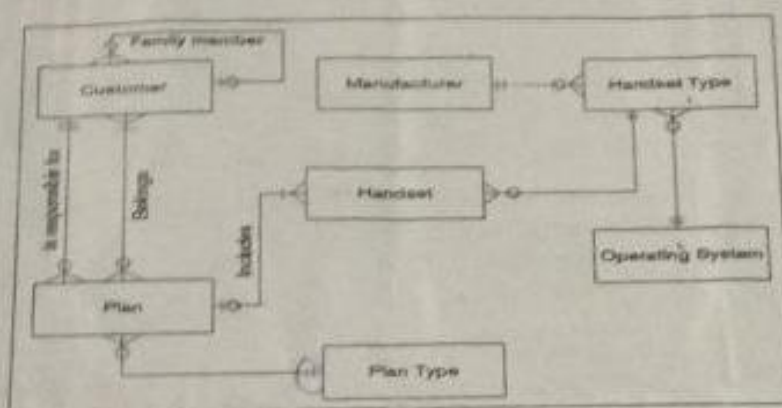


Figure 3: Customer Design

- (i) Can a customer have an unlimited number of plans?
 (ii) Can a customer exist without a plan?
 (iii) Is it possible to create a plan without knowing who the customer is?
 (iv) Does the operator want to limit the types of hand sets that can be linked to a specific plan type?
 (v) Is it possible to maintain data regarding a hand set without connecting it to a plan?
 (vi) Can a handset be associated with multiple plans?
 (vii) Assume a handset type exists that can utilize multiple operating systems. Could this situation be accommodated within the model included in Figure 3?
 (viii) Is the company able to track a manufacturer without maintaining information about its handsets?
 (ix) Can the same operating system be used on multiple handset types?
 (x) There are two relationships between Customer and Plan. Explain how they differ.
 (xi) Characterize the degree and the cardinalities of the relationship that connects Customer to itself. Explain its meaning.
 (xii) Is it possible to link a handset to a specific customer in a plan with multiple customers?
 (xiii) Can the company track a handset without identifying its operating system?

Mark 10/17

anything has been for me